SCOPE OF ACCREDITATION TO ISO/IEC 17025:2005

BRAUN INTERTEC CORPORATION
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CALIBRATION

Valid To: March 31, 2018 Certificate Number: 3940.01

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following calibrations[1]:

I. Dimensional

Parameter/Equipment	Range	CMC[2] (±)	Comments
Concrete Air Meter (Volumetric)	Up to 9 % of air in concrete	0.19 % of air in concrete	ASTM C173
Capping Jig (Alignment Device Perpendicularity Only)	(0 to 5)°	0.10°	ASTM C617
CBR Equipment –			ASTM D1883
Spacer	Up to 6 in	0.0018 in	
Weights	Up to 5 lb	0.0070 lbs	
Compactor Gyratory –			
Angle	Up to 125°	0.019°	AASHTO T312/T344
Height	Up to 115 mm	0.018 mm	
Rate	Up to 30 rpm	0.50 rpm	
Load	Up to 800 kpa	7.0 kpa	

Parameter/Equipment	Range	CMC[2, 3] (±)	Comments
Compactor Marshall or Proctor – Drop Weights Hammer Diameter	 Up to 18 in Up to 10 lb Up to 51 mm	 0.010 in 0.0070 lb 0.046 mm	 ASTM D6926/D698/D1557
Conical Mold and Tamper – Weight Diameters Height	 Up to 200 g Up to 100 mm Up to 75 mm	 0.31 g 0.046 mm 0.046 mm	 ASTM C128
FAA Cylinder – Volume	Up to 100 mL	0.13 mL	ASTM C1252
Flakiness Gauge – Openings	Up to 0.5 in	0.0018 in	BS-812, MnDOT 1223
Flat & Elongated Jig – Opening Ratio	Up to 2 in	0.0018 in	ASTM D4791
Flexural Strength Fixture – Length	 Up to 18 in	 0.010 in	 ASTM C78
LA Abrasion – Rate	Up to 35 rpm	0.16 rpm	ASTM C131-C535
Dial Indicator	Up to 4 in	0.00019 in	
Digital Indicator	Up to 0.5 in	0.00019 in	

Parameter/Equipment	Range	CMC² (±)	Comments
Liquid Limit Device and Groover – Cup Radius Dimensions Rate	 (52 to 56) mm Up to 150 mm Up to 2 rps	 0.39 mm 0.046 mm 0.16 rps	Rps = revolutions per second ASTM D4318
Marshall Breaking Head – Radius	Up to 2 in	0.0032 in	ASTM D6927
Molds Beam – Dimensions	Up to 18 in	0.010 in	ASTM C470
Molds CBR – Dimensions Volume	 Up to 120 mm Up to 3000 mL	 0.046 mm 3.6 mL	 ASTM D1883
Molds Cylinder – Dimensions	Up to 12 in	0.0018 in	ASTM C470
Molds Gyratory – Diameter	Up to 150 mm	0.017 mm	AASHTO T312
Molds Marshall – Dimensions	Up to 4 in	0.0018 in	ASTM D6926
Molds Mortar Cube – Dimensions	Up to 2 in	0.0018 in	ASTM C109
Molds Proctor – Dimensions Volume	 Up to 6 in Up to 3000 mL	 0.0018 in 3.6 mL	 ASTM D698/D1557

Parameter/Equipment	Range	CMC[2] (±)	Comments
Molds R-Value – Diameter Height	 Up to 4 in Up to 6 in	 0.00066 in 0.0018 in	 ASTM D2844
Soil Cement Molds – Dimensions	Up to 6 in	0.0018 in	ASTM D698
Steel Cylinder Molds – Dimensions	Up to 12 in	0.0018 in	ASTM C470
Neoprene Retaining Cups – Dimensions	Up to 6.5 in	0.0018 in	ASTM C1231
Proctor Straightedge – Beveled Thickness	Up to 0.25 in	0.0018 in	ASTM D698/D1557
Proving Ring – Indicated Load	Up to 1000 lb Up to 10 000 lb	1.3 lbs 5.7 lbs	ASTM D2166 ASTM D6927
Pycnometer – Gravimetric	Up to 250 mL Up to 500 mL Up to 4000 mL	0.29 mL 0.77 mL 4.7 mL	ASTM D854 ASTM C128 ASTM D2041
Sand Cone Bulk Density – Loose Density	Up to 100 lb/ft^3	0.22 lb/ft^3	ASTM D1556
Sand Cone and Plate – Weight to fill	Up to 2000 g	3.5 g	ASTM D1556

Parameter/Equipment	Range	CMC[2] (±)	Comments
Sand Equivalent – Plunger Weight Misc Dimensions	 Up to 1000 g Up to 400 mm	 1.3 g 0.046 mm	 ASTM D2419
Slump Cone and Rod – Diameters Heights/Lengths	 Up to 8 in Up to 24 in	 0.0018 in 0.010 in	 ASTM C143
Unit Weight Bucket – Volume	 Up to 15 000 mL	 17 mL	 ASTM C29/C138
Calipers – Linearity ID Jaws Depth Step Squareness	 Up to 12 in	 0.00031 in 0.00031 in 0.00031 in 0.00031 in	

II. Mechanical

Parameter/Equipment	Range	CMC[2] (±)	Comments
Concrete Air Meter (Pressure Method)	Up to 10 % of air in concrete	0.27 % of air in concrete	ASTM C231
Pressure	Up to 15 psi (10 to 100) psi	0.017 psi 0.45 psi	

Parameter/Equipment	Range	CMC[2] (±)	Comments
Electronic Balance – Field Scales General Purpose Analytical	 Up to 150 lb Up to 30 000 g Up to 5000 g Up to 200 g	 0.092 lb 0.32 g 0.023 g 1.2 mg	 Weights
Load Frame Concrete – Indicated Load	Up to 50 000 lb (50 000 to 500 000) lb	57 lb 1060 lb	ASTM E4
Load Frame Consolidation – Applied Load	Up to 2000 lb	1.3 lb	ASTM E4
Load Frame Direct Shear – Applied Load	Up to 2000 lb	1.3 lb	ASTM D3080
Load Frame Marshall Stability – Indicated Load	Up to 10 000 lb	5.7 lb	ASTM E4
Load Frame R-Value – Applied Load	Up to 10 000 lb	5.7 lb	ASTM D2844
Load Frame Triaxial – Applied Load	Up to 2000 lb	1.3 lb	ASTM D2850/D4767
Stressing Ram – Indicated Force	Up to 10 000 psi	85 psi	
Vacuum System – Absolute Pressure	(0 to 100) mmHg	2.3 mmHg	

III. Thermodynamics

Parameter/Equipment	Range	CMC[2] (±)	Comments
Temperature Indicators and Recorders	(-25 to 140) °C	0.28 °C	ASTM E644
Infrared Thermometers	(25 to 135) °C	0.65 °C	Blackbody target
Curing Room/Cabinet – Temperature Humidity	 (0 to 25) °C (0 to 75) % RH	 0.041 °C 3.9 % RH	 ASTM C511
Curing Tank – Temperature	 (0 to 25) °C	 0.041 °C	 ASTM C511
Evaporation – Oven	(0 to 50) mL/hr	0.015 mL/hr	ASTM C88
Temperature – Oven	(0 to 300) °F	2.4 °F	ASTM D2216
Humidity – Indicators and Recorders	(0 to 100) % RH	3.9 % RH	

IV. Time and Frequency

Parameter/Equipment	Range	CMC[2] (±)	Comments
Timer – Digital/Mechanical	Up to 3600 sec	0.16 sec	NIST publication 960-12 direct comparison

[1] This laboratory offers commercial calibration services.

[2] Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMCs represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of $k = 2$. The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

Accredited Laboratory

A2LA has accredited

BRAUN INTERTEC CORPORATION

Bloomington, MN

for technical competence in the field of

Calibration

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2005 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets any additional program requirements in the field of calibration. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (*refer to joint ISO-ILAC-IAF Communiqué dated 8 January 2009*).

AMERICAN ASSOCIATION FOR LABORATORY ACCREDITATION
CORPORATE SEAL 1978 DISTRICT OF COLUMBIA
A2LA

Presented this 16th day of March 2016.

Senior Director of Quality and Communications
For the Accreditation Council
Certificate Number 3940.01
Valid to March 31, 2018

For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.